Node Vortex

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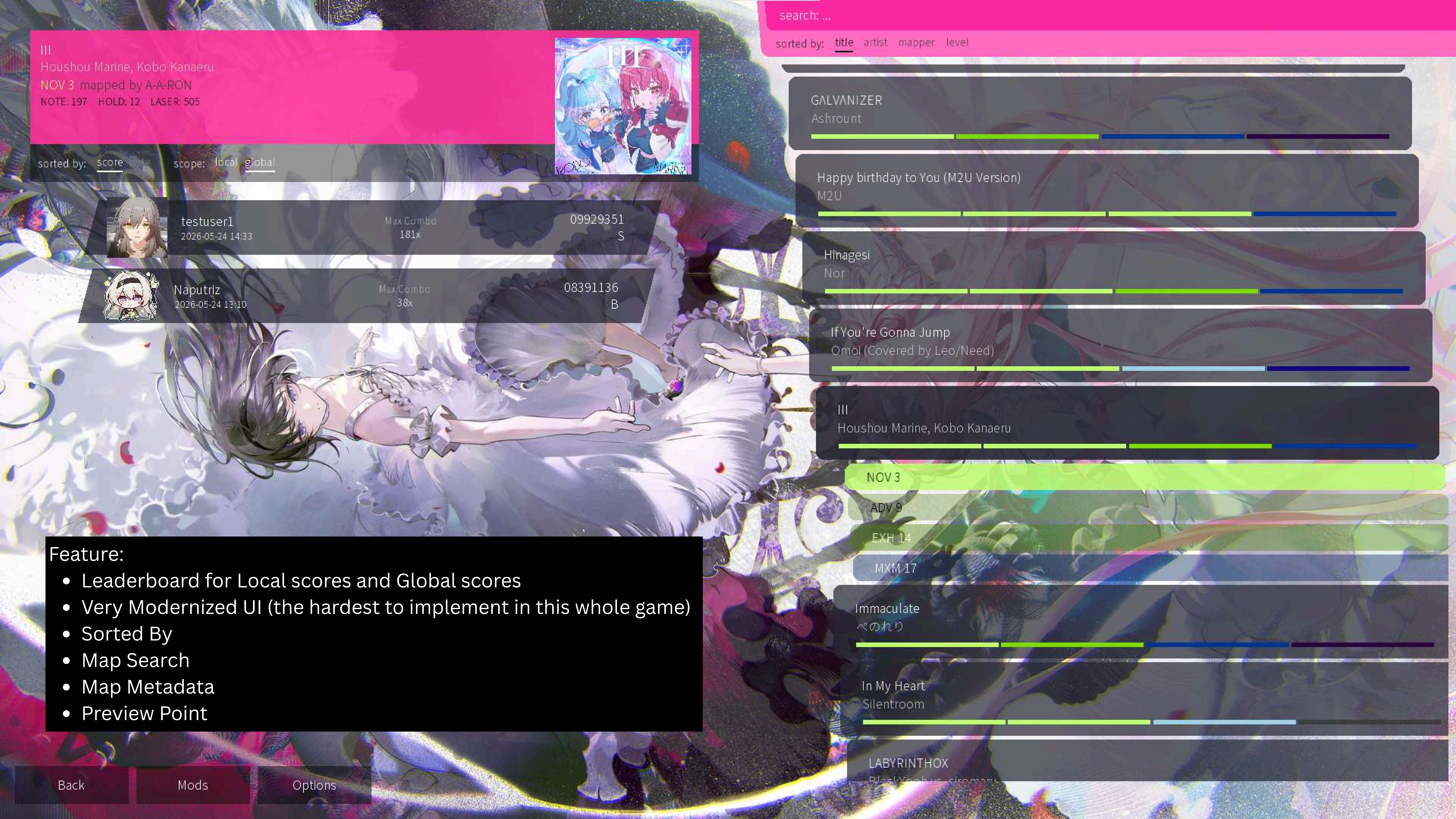
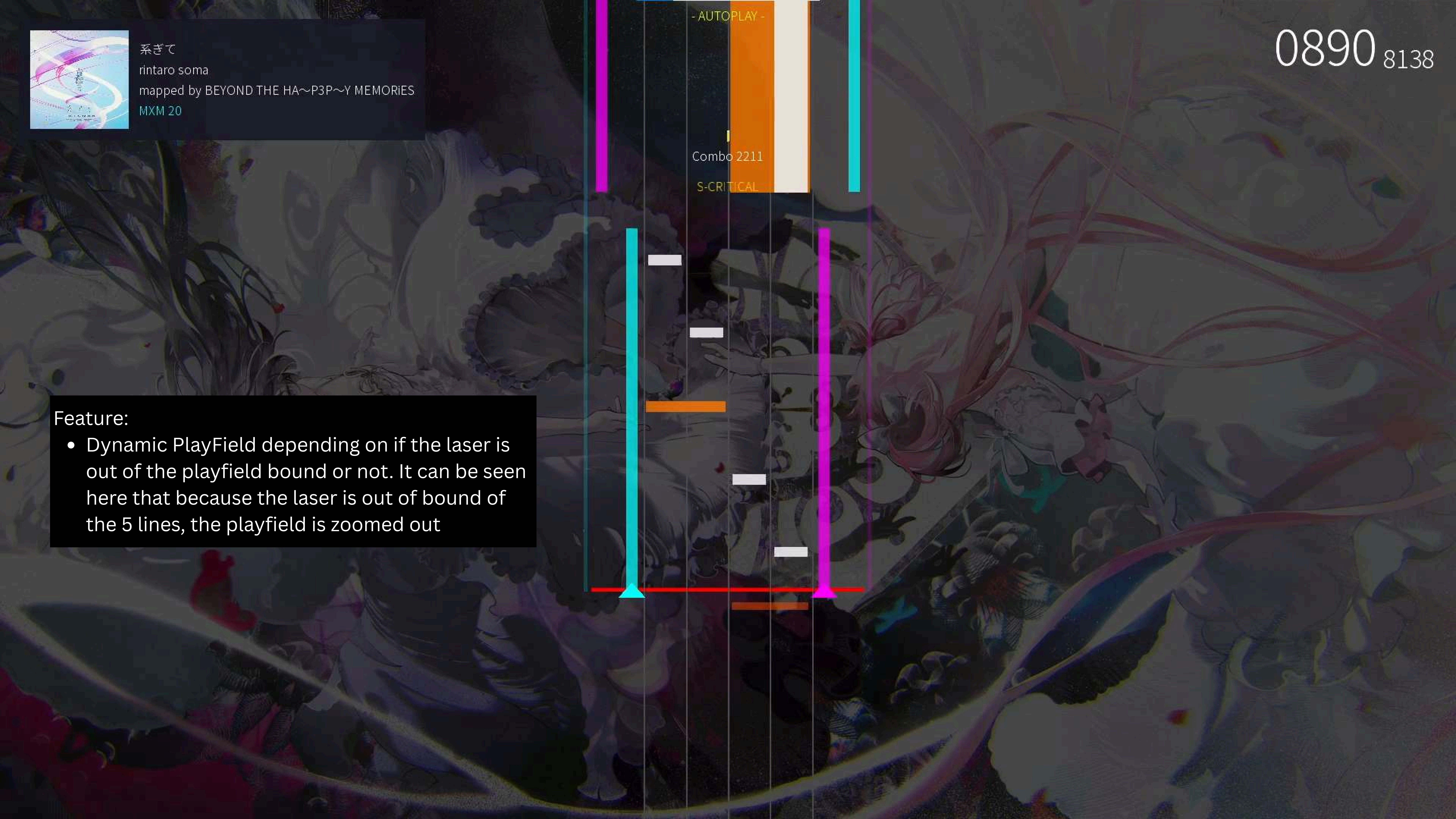
START
EXIT



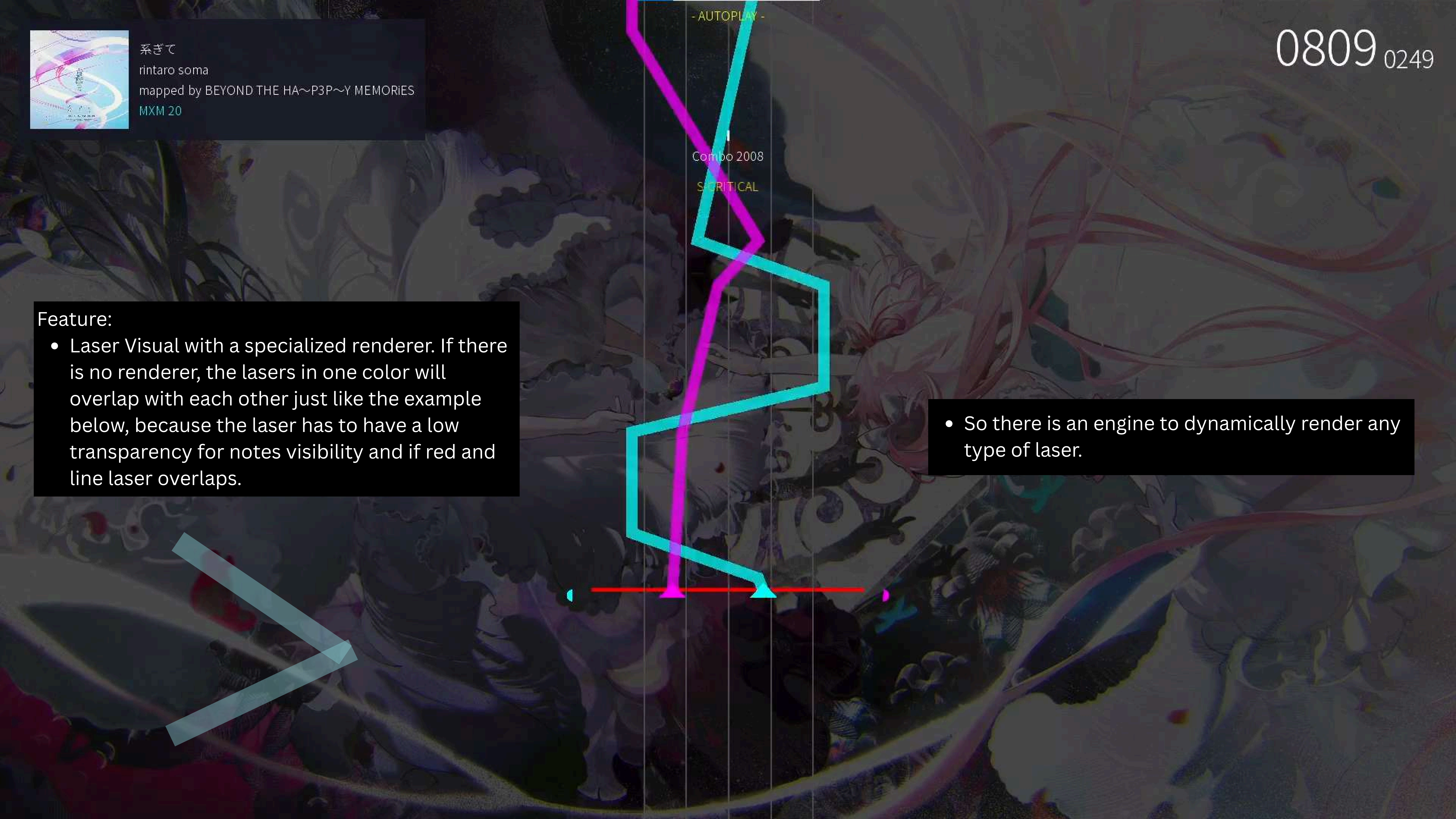
Feature:

- Leaderboard for Local scores and Global scores
- Very Modernized UI (the hardest to implement in this whole game)
- Sorted By
- Map Search
- Map Metadata
- Preview Point



Feature:

- Dynamic PlayField depending on if the laser is out of the playfield bound or not. It can be seen here that because the laser is out of bound of the 5 lines, the playfield is zoomed out



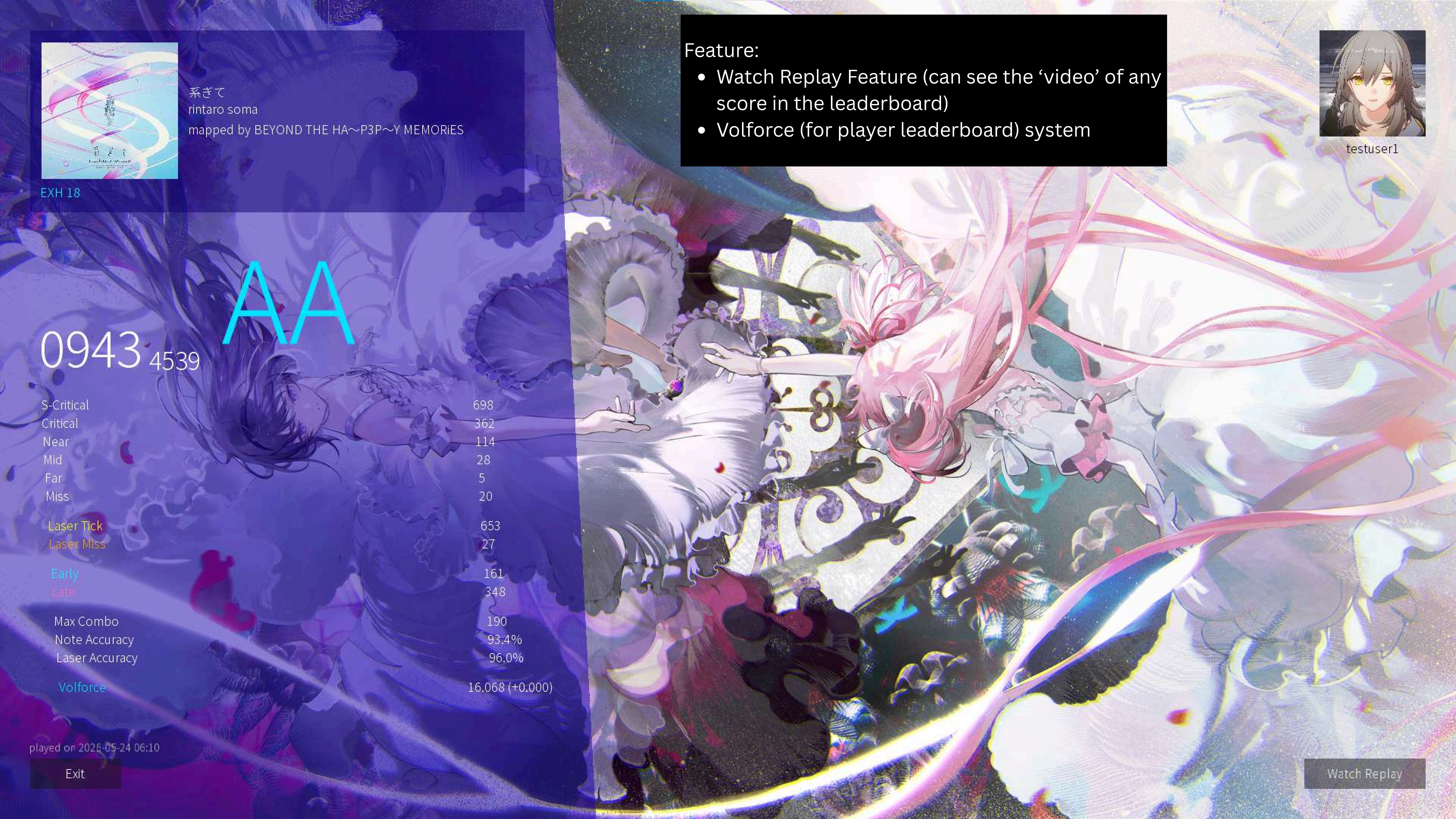
Feature:

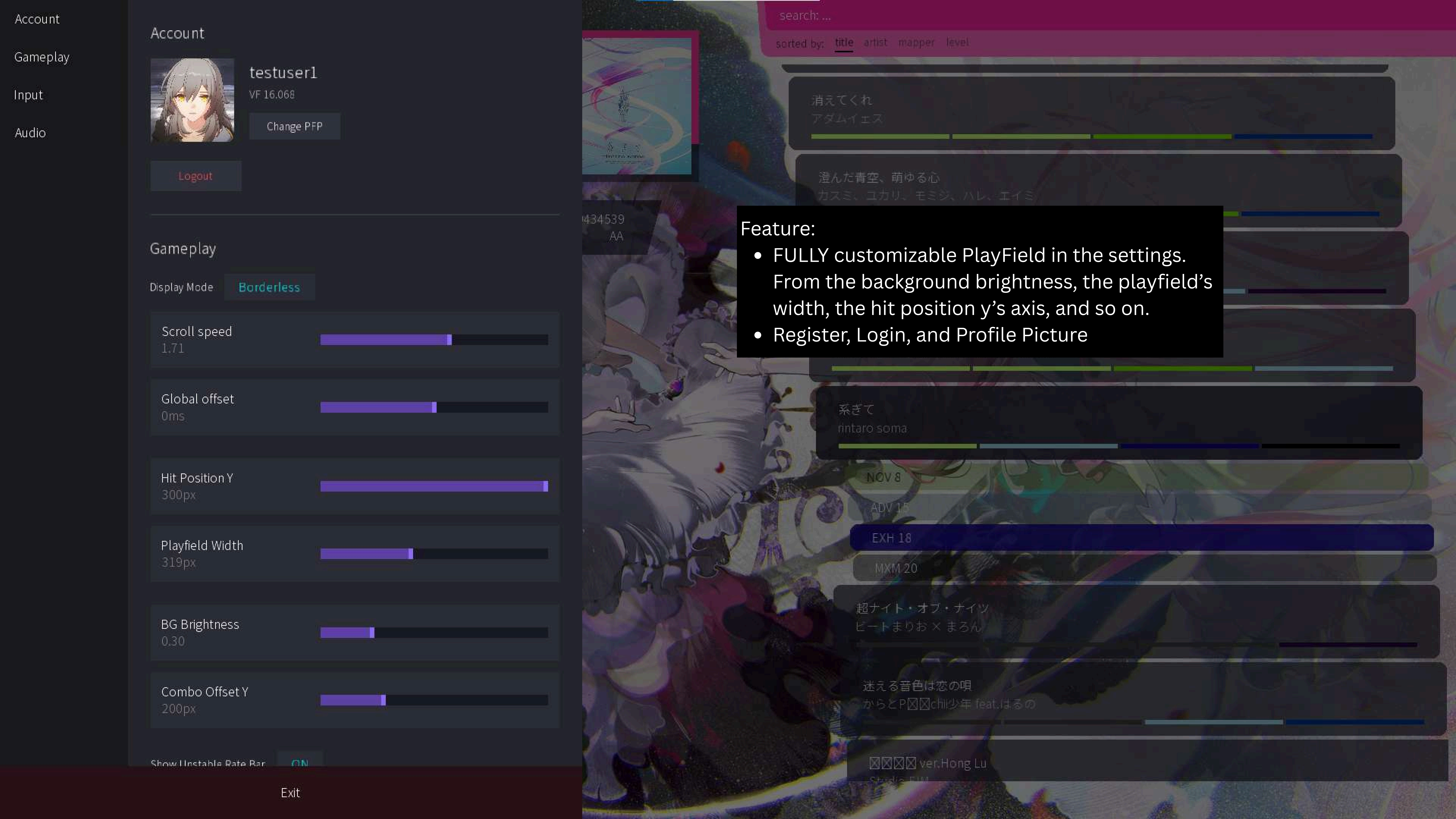
- Laser Visual with a specialized renderer. If there is no renderer, the lasers in one color will overlap with each other just like the example below, because the laser has to have a low transparency for notes visibility and if red and line laser overlaps.

- So there is an engine to dynamically render any type of laser.

Feature:

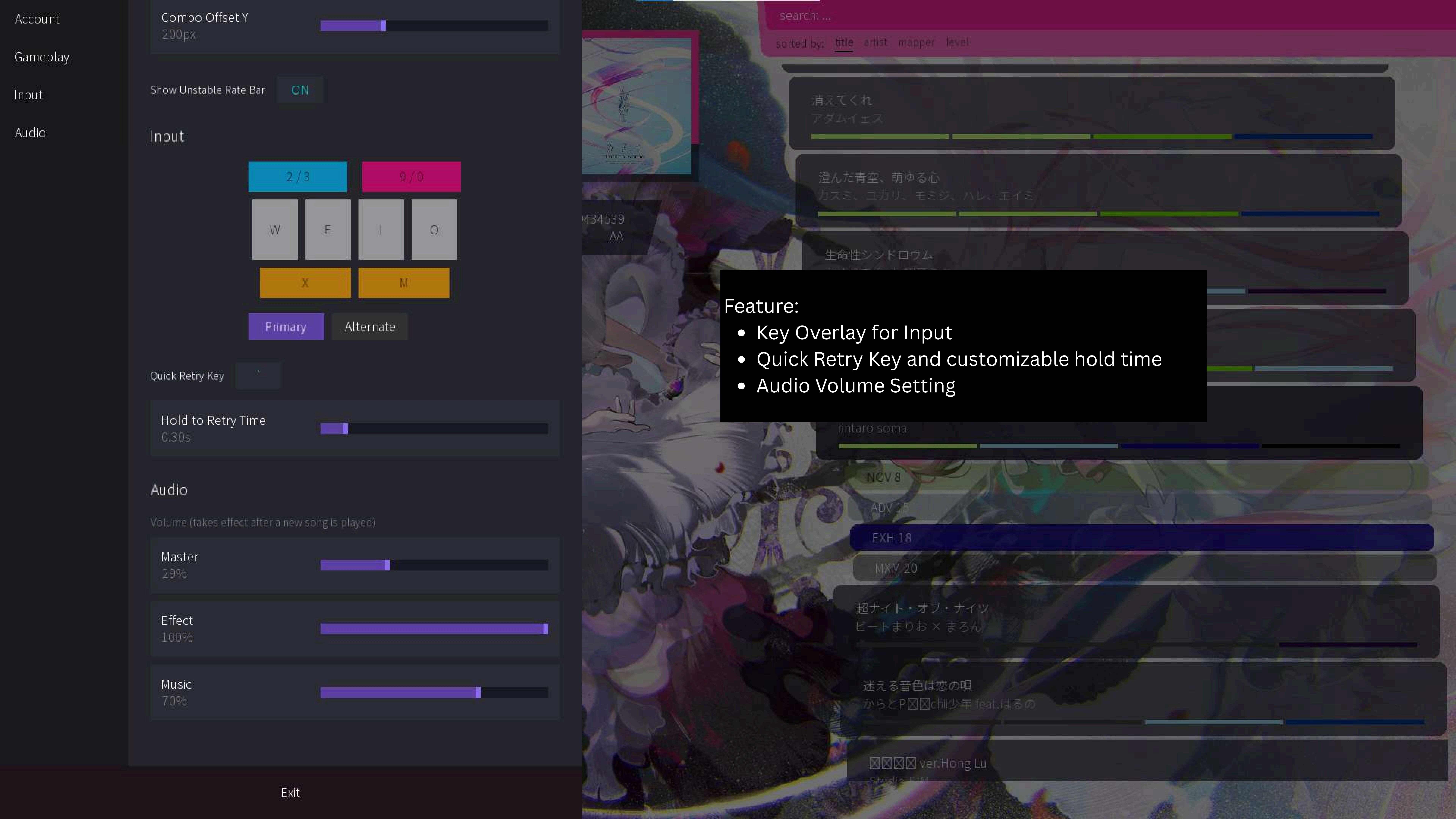
- Watch Replay Feature (can see the ‘video’ of any score in the leaderboard)
- Volforce (for player leaderboard) system

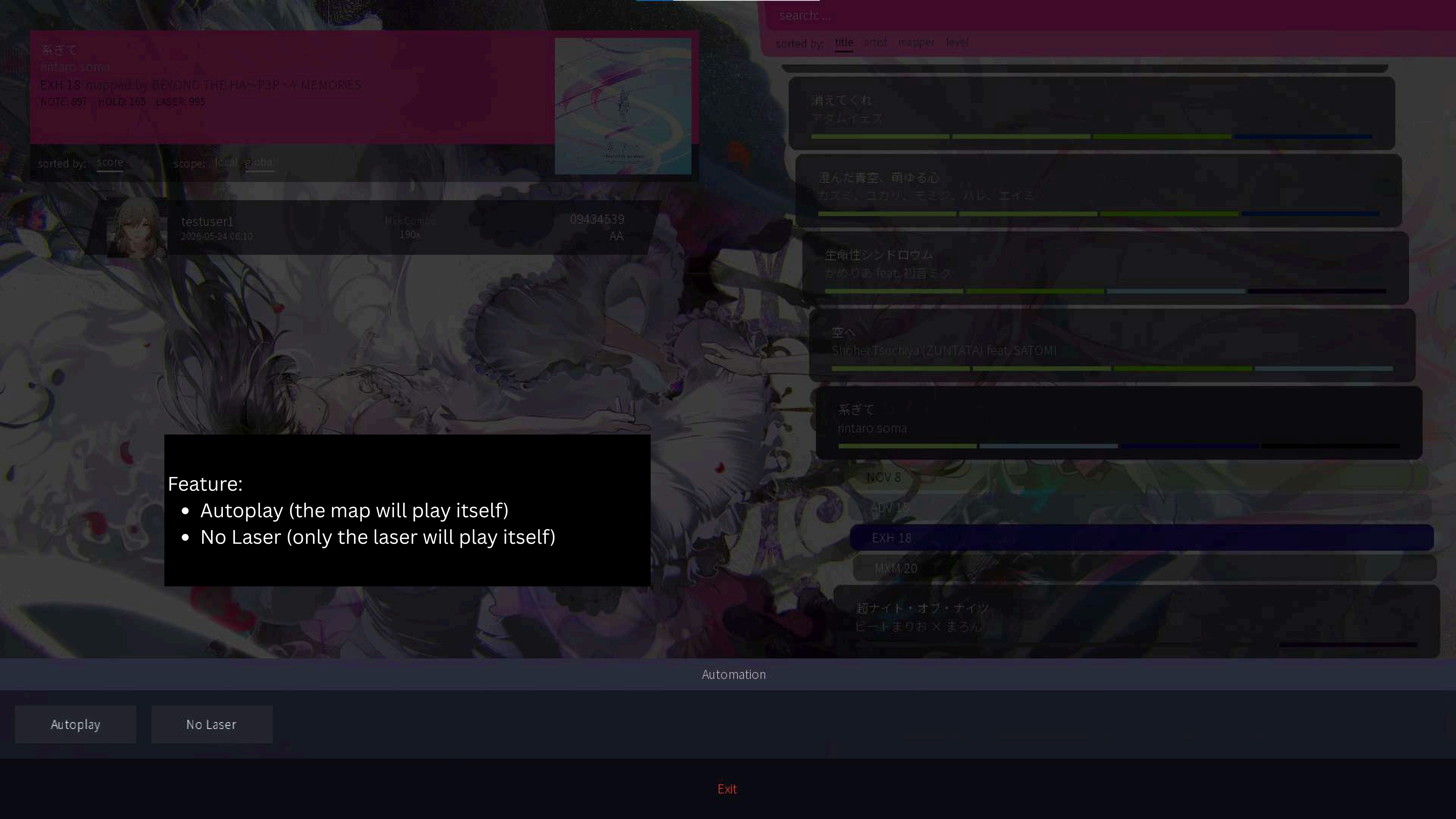




Feature:

- FULLY customizable PlayField in the settings. From the background brightness, the playfield’s width, the hit position y’s axis, and so on.
- Register, Login, and Profile Picture


Account
Gameplay
Input
Audio
Combo Offset Y
200px
Show Unstable Rate Bar
ON
Input
2 / 3
9 / 0
W
E
I
O
X
M
Primary
Alternate
Quick Retry Key
Hold to Retry Time
0.30s
Audio
Volume (takes effect after a new song is played)
Master
29%
Effect
100%
Music
70%
Exit
search: ...
sorted by:
title
artist
mapper
level
消えてくれ
アダムイェス
澄んだ青空、萌ゆる心
生命性シンドロウム
434539
AA
rintaro soma
NOV 8
ADV 15
EXH 18
MXM 20
超ナイト・オブ・ナイツ
ビートまりお × まろん
ver.Hong Lu
Feature:
Key Overlay for Input
Quick Retry Key and customizable hold time
Audio Volume Setting


search: ...
sorted by: title artist mapper level
sorted by: score
scope: local global
testuser1
09434539
AA
消えてくれ
生命性シンドロウム
空へ
Shohei Tsuchiya (ZUNTATA) feat. SATOMI
rintaro soma
NOV 8
EXH 18
MXM 20
超ナイト・オブ・ナイツ
Feature:
• Autoplay (the map will play itself)
• No Laser (only the laser will play itself)
Automation
Autoplay
No Laser
Exit

6 Design Pattern

- Strategy Pattern:
  - e.g. ScoreManager.java

```java
private final StrictJudgment judgmentStrategy;  2 usages

public ScoreManager(StrictJudgment strategy) { this.judgmentStrategy = strategy; }

// split Notes and Releases for the Max Score math
public void setMaxPossibleScore(int totalNotes, int totalReleases, int totalLaserTicks) {  1 usage  arkaanp
    this.maxHitScore = (totalNotes * 3.0f) + (totalReleases * 3.0f) + (totalLaserTicks * 3.0f);
    this.maxLaserTicks = totalLaserTicks;
}

public void onHit(float diffMs, String type) {  5 usages  arkaanp
    StrictJudgment.JudgmentResult result = judgmentStrategy.evaluateJudgment(diffMs, type);
    StatCategory stats = type.equals("RELEASE") ? releaseStats : noteStats;

    if (!result.tier.equals("MISS")) {
        combo++;
        if (combo > maxCombo) maxCombo = combo;

        // --- Record the hit for the UR Bar (Ignore lasers) ---
        if (!type.equals("LASER")) {
            recentHits.add(new HitMarker(diffMs, result.tier));
        }
    } else {
        combo = 0;
    }

    // Only update UI text for Notes/Releases, and format it clearly (e.g., "NEAR EARLY")
    if (!result.tier.equals("S-CRITICAL") && !result.tier.equals("MISS")) {
```

The `JudgmentStrategy` interface allows us to swap different scoring algorithms (like `StrictJudgment`) into the `ScoreManager`. This lets us change how "strict" the timing is without rewriting the score logic.

6 Design Pattern

- Command Pattern:
  - e.g. ReplayData.java & PlayScreen.java

```java
public class ReplayData {  13 usages  arkaanp
    public String difficulty = "";  1 usage
    public int finalScore = 0;  1 usage
    public long timestamp = 0;  1 usage

    // A chronological list of every single button press/release
    public Array<InputEvent> events = new Array<>();  8 usages

    public static class InputEvent {  arkaanp
        public float audioTimeMs;  2 usages
        public String inputType; // "BT1", "BT2", "FX_L", "LASER_L", etc.  2 usages
        public boolean isPressed; // true = Pressed down, false = Released  2 usages

        // Empty constructor needed for JSON parsing
        public InputEvent() {}  arkaanp

        public InputEvent(float audioTimeMs, String inputType, boolean isPressed) {  arkaanp
            this.audioTimeMs = audioTimeMs;
            this.inputType = inputType;
            this.isPressed = isPressed;
        }
    }
}
```

```java
    continueBtn.addListener(new ClickListener() { @Override public void clicked(InputEvent event, float x, float y) { resumeGame(); } });
    retryBtn.addListener((ClickListener) clicked(event, x, y) → {  arkaanp
            if (music != null) music.stop();
            game.setScreen(new PlayScreen(game, mapFilePath,  replayTimestampToLoad: 0L,  isRetry: true));
    });
    exitBtn.addListener((ClickListener) clicked(event, x, y) → {
            if (music != null) music.stop();
            if (game.songSelectScreen != null) {
                game.songSelectScreen.dispose();
            }
            game.songSelectScreen = new SongSelectScreen(game,  mainMenuMusic: null,  mapFilePath,  slideInFromRight: true);
            game.setScreen(game.songSelectScreen);
    });
}
```

The `InputEvent` class encapsulates key presses as objects. These "commands" are stored in a list and then "executed" during replays to mimic the player's exact movements.

6 Design Pattern

- Object Pool Pattern:
  - e.g. Note.java & PlayScreen.java

```java
import com.badlogic.gdx.graphics.glutils.ShapeRenderer;
import com.badlogic.gdx.utils.Pool;

public class Note implements Pool.Poolable {  19 usages  new *
    private final float NOTE_THICKNESS = 20f;  2 usages
    public int lane;  11 usages
    public float startTime;  8 usages
    public float endTime;  5 usages
    public boolean isHold;  14 usages
    public boolean isActive;  3 usages

    public boolean wasHeadHit;  11 usages
    public boolean isCompleted;  11 usages
    public boolean isMissed;  9 usages

    public float y;

    public Note() {}  1 usage  new *

    public void init(int lane, float startTime, float endTime, boolean isHold) {  1 usage  new *
        this.lane = lane;
        this.startTime = startTime;
        this.endTime = endTime;
        this.isHold = isHold;

        this.isActive = true;
        this.wasHeadHit = false;
        this.isCompleted = false;
        this.isMissed = false;
    }
```

```java
// Object Pool for Notes
private Array<Note> activeNotes = new Array<>();  10 usages
private final Pool<Note> notePool = new Pool<Note>() {  2 usages  arkaanp
    @Override protected Note newObject() { return new Note(); }  arkaanp
};

private Note createNote(int lane, float start, float end, String type) {  1 usage  arkaanp
    Note note = notePool.obtain();
    note.init(lane, start, end, "HOLD".equals(type));
    return note;
}
```

To prevent lag (Garbage Collection), the game doesn't delete `Note` objects. Instead, it uses a `Pool<Note>` to recycle them. When a note goes off-screen, it is "reset" and put back in the pool for the next song.

6 Design Pattern

- Observer Pattern:
  - e.g. NetworkManager.java & TextureLoader.java

```java
public interface NetworkCallback<T> {  17 usages  11 implementations  arkaanp
    void onSuccess(T result);  8 usages  11 implementations  arkaanp
    void onError(String message);  23 usages  11 implementations  arkaanp
}
```

```java
public static void register(String username, String password, final NetworkCallback<String> callback) {  1 usage
    Net.HttpRequest request = new Net.HttpRequest(Net.HttpMethods.POST);
    request.setUrl(BASE_URL + "/auth/register");
    request.setHeader( name: "Content-Type",  value: "application/json");
    request.setTimeOut(20000);
```

```java
public interface TextureCallback {  2 usages  1 implementation  arkaanp
    void onLoaded(Texture texture);  2 usages  1 implementation  arkaanp
    void onError(Throwable t);  4 usages  1 implementation  arkaanp
}

public static void loadTexture(String url, final TextureCallback callback) {
    url = transformCloudinaryUrl(url);
    if (url == null || url.isEmpty()) {
        callback.onError(new Exception("Empty URL"));
        return;
    }
```

The `NetworkCallback<T>` and `TextureCallback` interfaces act as observers. The UI "subscribes" to the results, and the managers "notify" the UI once the profile picture or score data has finished loading.

6 Design Pattern

- State Pattern:
  - e.g. LaserCursor.java

```java
public class LaserManager {  3 usages  arkaanp
    public void updateCursor(LaserCursor cursor, Array<Beatmap.LaserSequence> laserData, float currentTime, float delta,
                }

                // --- APPLY FINAL STATE ---
                if (wrongInput) {
                    cursor.isMissed = true;
                    cursor.missedTimer += delta * 1000f;
                    cursor.setState(new com.nodevoltex.game.patterns.FreeState());

                    // They stay physically stranded wherever they fell off so they can see their mistake

                    if (cursor.missedTimer >= graceWindowMs && !cursor.hasComboBroken) {
                        scoreManager.onMiss( type: "LASER");
                        cursor.hasComboBroken = true;
                    }
                } else {
                    cursor.isMissed = false;
                    cursor.missedTimer = 0f;
                    cursor.hasComboBroken = false;
                    cursor.setState(new LockedState());
```

- CursorState
- FreeState
- GameArchitec
- LockedState

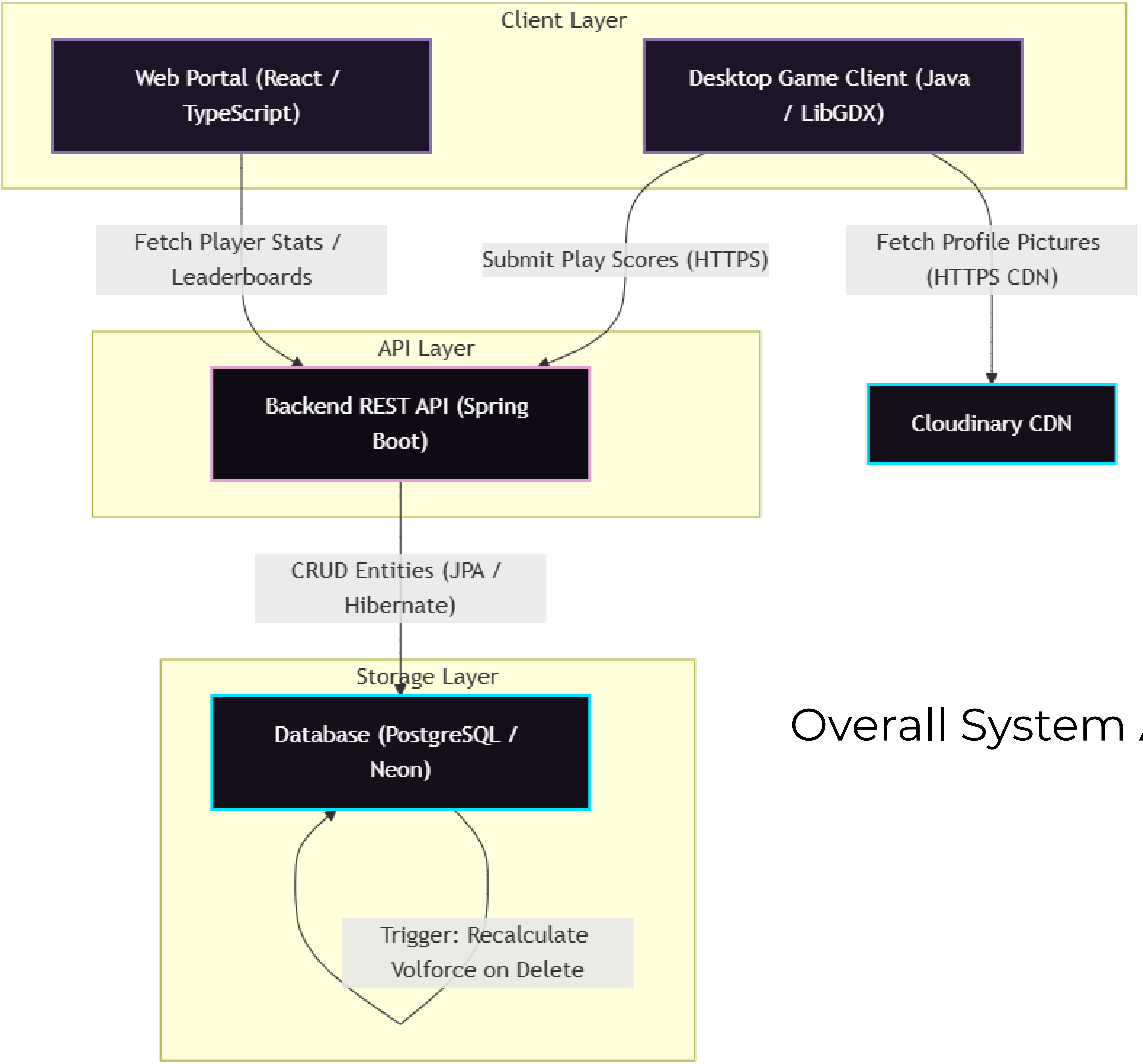
The laser cursor uses the State pattern to switch between `FreeState` (moving freely) and `LockedState` (attached to a slider). The behavior of the cursor changes completely depending on which state object is active.

6 Design Pattern

- Singleton Pattern:
  - e.g. SettingsManager.java (and Backend Services)

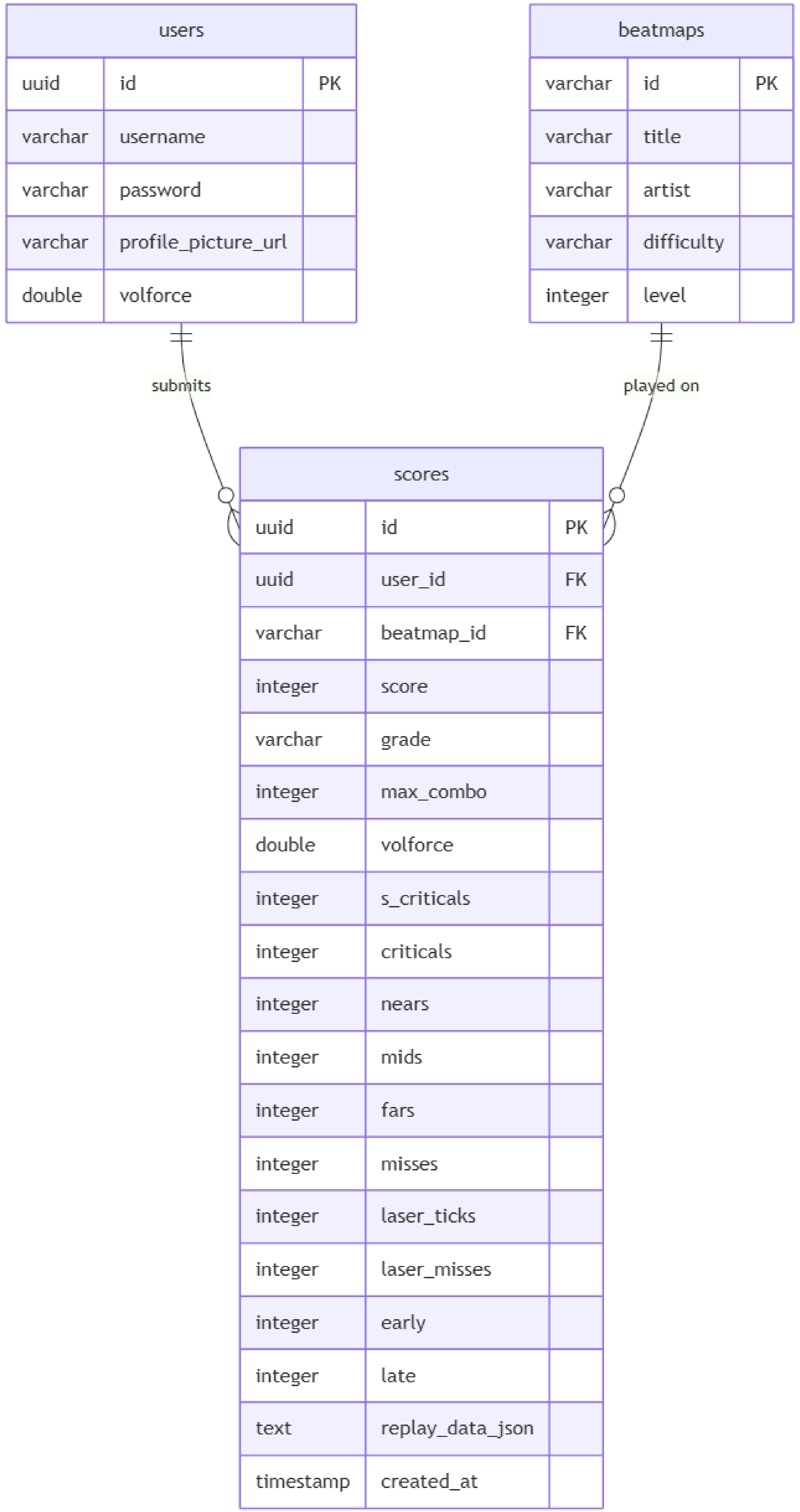
```java
package com.nodevoltex.game.managers;

import ...

public class SettingsManager {  arkaanp
    private static Preferences prefs;  3 usages

    private static Preferences getPrefs() {  60 usages  arkaanp
        if (prefs == null) prefs = Gdx.app.getPreferences( s: "NodeVoltexSettings");
        return prefs;
    }

    // --- DEFAULTS APPLIED HERE ---
    public static float getScrollSpeed() { return getPrefs().getFloat( s: "scrollSpeed",  v: 1.2f); }  3 usages
    public static float getGlobalOffset() { return getPrefs().getFloat( s: "globalOffset",  v: 0f); }  4 usages

    // Master Volume defaulted to 30%
    public static float getMasterVolume() { return getPrefs().getFloat( s: "masterVolume",  v: 0.3f); }  9 usage
    public static float getMusicVolume() { return getPrefs().getFloat( s: "musicVolume",  v: 0.8f); }  8 usages
    public static float getEffectVolume() { return getPrefs().getFloat( s: "effectVolume",  v: 1.0f); }  4 usage

    // --- Playfield Customization
```

The `SettingsManager` provides a single, global point of access for all game configurations. In the Backend, Spring Boot manages your services (like `UserService`) as singletons automatically.



Overall System Architecture

Database Entity Relationship Diagram (ERD)



High Score Submission Sequence